How to use this interactive PDF: Click **Show / Hide Answer** under any question to reveal or hide its answer. This requires **Adobe Acrobat Reader** (free). In other viewers the buttons may not work — use your viewer's **Layers** panel, or the **Reveal All** button below.

Reveal All   Hide All

CCNA Subnetting Drill Sheet

50+ Problems with Full Worked Answers — From Baby Steps to Exam Speed

> **Why this exists:** Subnetting shows up ALL OVER the CCNA exam, and you have to do it **fast** (about 60–90 seconds per question). Reading about it isn't enough — you have to drill it until it's automatic, like times tables. This sheet gives you 50+ problems, grouped from easy to hard, each with a **step-by-step answer**.
>
> **How to use it:** Cover the answer with your hand or a piece of paper. Work the problem on scratch paper FIRST. Then check. Do **5–10 problems every single day**. In two weeks you'll be fast.

The Toolkit (Everything You Need on One Page)

The Two Magic Charts

Chart 1 — Bit values in one octet (memorize left to right):

```
128   64   32   16   8   4   2   1
```

Chart 2 — The subnet "cheat table" (memorize this cold):

```
CIDR   Mask (last octet)   Block Size   Hosts (usable)
/24       0                   256          254
/25       128                 128          126
/26       192                  64           62
/27       224                  32           30
/28       240                  16           14
/29       248                   8            6
/30       252                   4            2
```

> Notice the pattern: **Block size doubles as you go up** (4, 8, 16, 32, 64, 128, 256), and **Mask value** = 256 − block size.

The 5 Facts You Compute for Any Subnet

For any IP + mask, you can always find these 5 things:

1. **Block size** = 256 − (the interesting octet of the mask).
2. **Network address** = the block start the IP falls into (all host bits 0).
3. **Broadcast address** = one below the next network (all host bits 1).
4. **First usable host** = network + 1.
5. **Last usable host** = broadcast − 1.

The Formulas

```
Number of hosts per subnet = 2^(host bits) − 2
Number of subnets          = 2^(borrowed bits)
Host bits                  = 32 − CIDR
```

The 3-Step "Where Does This IP Live?" Method

1. Find the **block size** (256 − interesting mask octet).
2. Count up in the interesting octet by the block size (0, block, 2×block…) until you pass the IP's value. The one **just below or equal** is the **network**.
3. **Broadcast** = next network − 1. Hosts sit between them.

SECTION 1 — Warm-Up: Masks, CIDR & Host Counts

(Goal: instant recall. Aim for under 15 seconds each.)

Q1. Convert `/26` to a dotted-decimal mask.

Show / Hide Answer

Q2. Convert `255.255.255.240` to CIDR.

Show / Hide Answer

Q3. Convert `/19` to a dotted-decimal mask.

Show / Hide Answer

Q4. How many **usable hosts** are in a `/27`?

Show / Hide Answer

Q5. How many **usable hosts** are in a `/30`?

Show / Hide Answer

Q6. How many **usable hosts** are in a `/22`?

Show / Hide Answer

Q7. Convert binary `11000000` to decimal.

Show / Hide Answer

Q8. Convert decimal `224` to binary.

Show / Hide Answer

Q9. What is the **block size** of a `/29`?

Show / Hide Answer

Q10. A mask ends in `.248`. What CIDR is that, and how many hosts?

Show / Hide Answer

SECTION 2 — Find the Network & Broadcast Address

(The bread-and-butter skill. Aim for under 60 seconds each.)

Q11. Given `192.168.1.77 /26`, find the network and broadcast address.

Show / Hide Answer

Q12. Given `10.0.0.200 /27`, find the network and broadcast.

Show / Hide Answer

Q13. Given `172.16.5.130 /25`, find the network and broadcast.

Show / Hide Answer

Q14. Given `192.168.10.44 /28`, find the network and broadcast.

Show / Hide Answer

Q15. Given `10.10.10.10 /29`, find the network and broadcast.

Show / Hide Answer

Q16. Given `192.168.1.100 /30`, find the network and broadcast.

Show / Hide Answer

Q17. Given `172.16.99.200 /21`, find the network and broadcast. (Now the interesting octet is the THIRD one!)

Show / Hide Answer

Q18. Given `10.50.150.7 /20`, find the network and broadcast.

Show / Hide Answer

Q19. Given `192.168.200.55 /18`, find the network and broadcast.

Show / Hide Answer

Q20. Given `10.1.1.1 /8`, find the network and broadcast.

Show / Hide Answer

SECTION 3 — Find the Usable Host Range

(Network + 1 to Broadcast − 1. Put it all together.)

Q21. For `192.168.1.77 /26`, list the **first and last usable host**.

Show / Hide Answer

Q22. For `172.16.5.130 /25`, list the first and last usable host.

Show / Hide Answer

Q23. For `10.10.10.10 /29`, list the first and last usable host.

Show / Hide Answer

Q24. For `192.168.1.100 /30`, list the first and last usable host.

Show / Hide Answer

Q25. For `172.16.99.200 /21`, list the first and last usable host.

Show / Hide Answer

Q26. Is `192.168.1.64` a **usable host** address in the `192.168.1.0 /26` scheme? Why?

Show / Hide Answer

Q27. Is `192.168.1.191` usable in a `/26` scheme? Why?

Show / Hide Answer

Q28. Two PCs: `192.168.1.70 /26` and `192.168.1.130 /26`. Can they talk **directly** (same subnet)?

Show / Hide Answer

Q29. Two PCs: `10.0.0.33 /27` and `10.0.0.60 /27`. Same subnet?

Show / Hide Answer

Q30. A device is configured `192.168.5.31 /27`. What's wrong?

Show / Hide Answer

SECTION 4 — "How Many Subnets / Hosts Do I Need?" (Design)

(These are the word problems. Read carefully!)

Q31. You have `192.168.1.0 /24` and need **4 equal subnets**. What mask and how many hosts each?

Show / Hide Answer

Q32. You have `192.168.1.0 /24` and need **at least 50 hosts per subnet**, as many subnets as possible. What mask?

Show / Hide Answer

Q33. You need **at least 500 hosts** in one subnet. What is the smallest mask (largest CIDR number) that works?

Show / Hide Answer

Q34. You need **exactly 6 subnets** from `172.16.0.0 /16`, each as large as possible. What mask?

Show / Hide Answer

Q35. How many `/30` subnets can you make from a single `/24`?

Show / Hide Answer

Q36. How many `/26` subnets fit in a `/22`?

Show / Hide Answer

Q37. A company needs 200 hosts in one LAN. Will a `/25` work? What about a single `/24`?

Show / Hide Answer

Q38. You need a subnet for a **point-to-point link between two routers** with no wasted addresses. What mask?

Show / Hide Answer

Q39. From `10.0.0.0 /24`, you subnet into `/28`s. What is the **network address of the 5th subnet**?

Show / Hide Answer

Q40. From `192.168.1.0 /24` subnetted into `/27`s, what is the **broadcast of the 3rd subnet**?

Show / Hide Answer

SECTION 5 — VLSM (Variable Length Subnet Masks)

(Different-sized subnets from one block. The GOLDEN RULE: always assign the biggest subnet first!)

Q41. From `192.168.1.0 /24`, create subnets for: Sales (100 hosts), IT (50 hosts), HR (25 hosts), and a WAN link (2 hosts). Give the subnet + mask for each.

Show / Hide Answer

Q42. From `10.0.0.0 /24`, you need three subnets: A (60 hosts), B (12 hosts), C (2 hosts). Provide subnet + mask for each.

Show / Hide Answer

Q43. In Q41, what goes wrong if you assign the **WAN /30 first** at 192.168.1.0, then try to fit Sales?

Show / Hide Answer

Q44. From `172.16.0.0 /22`, allocate: Building1 (500 hosts), Building2 (250 hosts), Building3 (100 hosts). Provide subnet + mask for each.

Show / Hide Answer

Q45. You subnetted `192.168.4.0 /24` with VLSM and used up to `192.168.4.223`. A new team needs **20 hosts**. What subnet + mask do you give them, and does it fit?

Show / Hide Answer

SECTION 6 — Rapid-Fire Mixed Drills (Exam Simulation)

(No hints. Do these on a timer — target under 60 seconds each.)

Q46. `192.168.3.194 /26` → network, broadcast, first, last?

Show / Hide Answer

Q47. `10.20.30.40 /28` → network, broadcast, first, last?

Show / Hide Answer

Q48. `172.31.100.100 /26` → network, broadcast?

Show / Hide Answer

Q49. `192.168.15.15 /29` → network, broadcast, first, last?

Show / Hide Answer

Q50. `10.0.5.130 /23` → network, broadcast? (Third-octet subnetting!)

Show / Hide Answer

Q51. `192.168.1.222 /27` → is it a host, network, or broadcast?

Show / Hide Answer

Q52. `172.16.14.0 /23` → how many usable hosts, and what's the broadcast?

Show / Hide Answer

Q53. `10.10.10.100 /25` → which subnet (network address)?

Show / Hide Answer

Q54. `192.168.100.100 /30` → network, broadcast, both usable hosts?

Show / Hide Answer

Q55. You see `255.255.255.252`. How many of these subnets fit in a `/24`, and what are they good for?

Show / Hide Answer

SECTION 7 — Boss Level: Full Scenario Problems

(These mimic the harder simulation-style exam questions. Take your time.)

Q56. A router interface is configured with `192.168.1.65 /26`. A host on that LAN is set to `192.168.1.130 /26` with gateway `192.168.1.65`. The host can't reach the internet. What's the problem and the fix?

Show / Hide Answer

Q57. You're given `192.168.50.0 /24` and must support **6 subnets**, each needing **25 usable hosts**. Give the mask and list all 6 network addresses.

Show / Hide Answer

Q58. `10.1.1.0 /24` needs these, biggest-first (VLSM): Subnet A 120 hosts, B 60 hosts, C 30 hosts, D 12 hosts, plus two /30 WAN links (E, F). List each subnet + mask + range.

Show / Hide Answer

Q59. Summarize (aggregate) these four networks into ONE route: `172.16.0.0/24, 172.16.1.0/24, 172.16.2.0/24, 172.16.3.0/24`.

Show / Hide Answer

Q60. Summarize `192.168.8.0/24` through `192.168.15.0/24` into one route.

Show / Hide Answer

The One-Page Speed Method (Print This!)

When you see any "IP + CIDR" question, do this in your head, every time:

```
STEP 1: Which octet is "interesting"?
        /1–/8  → 1st octet
        /9–/16 → 2nd octet
        /17–/24→ 3rd octet
        /25–/32→ 4th octet

STEP 2: Block size = 256 − (mask value in that octet)

STEP 3: Count 0, block, 2×block... until you pass the IP.
        The one at/just-below = NETWORK.

STEP 4: Broadcast = next network − 1
        First host = network + 1
        Last host  = broadcast − 1

STEP 5: Usable hosts = 2^(32−CIDR) − 2
```

Final Tips

1. **Memorize the cheat table** (/24–/30 with block sizes and host counts). It saves you 30 seconds every question.
2. **Always find the block for BOTH addresses** when asked "same subnet?" — don't eyeball it.
3. **Watch out for network & broadcast addresses** disguised as host answers (a favorite trick).
4. **VLSM golden rule:** biggest subnet first, every time.
5. **Do 10 problems a day.** Speed comes only from reps. In two weeks this will feel easy.

You've got this. Now go drill!

— End of Subnetting Drill Sheet —